1.5 Connecting to a network

In this lesson

You will learn:

- how to connect to a network
- basic troubleshooting methods for network connections.



Types of network connection

There are three main ways to connect to a network.

1 **Wired** Your computer is connected to a data point on a wall with a cable. You can only connect to a network if you have a network point nearby. You cannot move your computer. Wired connections are used for desktop computers and sometimes for laptops.

2 **Wireless (Wi-Fi)** A cable isn't needed for a wireless connection. Signals are sent between your computer and a wireless access point (WAP). Lesson 1.1 has more information about WAPs.

Laptop computers, tablets and smartphones connect to networks using Wi-Fi. Using Wi-Fi means you can move around without easily losing your connection.

Wireless networks are sometimes available in public places like restaurants or shops. A WAP in a public place is called a **hotspot**.

3 **Mobile phone** Smartphones can connect to the internet using a mobile phone connection. A connection can be made anywhere there is a phone signal.

How to connect to a network

School network

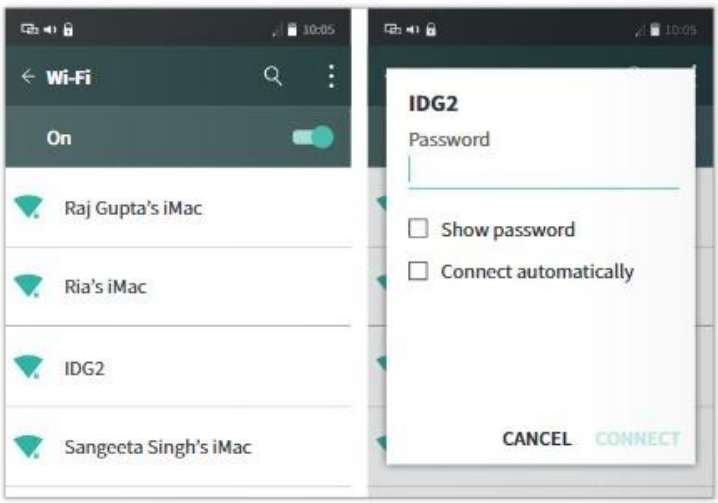
Everyone who is allowed to use the school network is given a username and password. You use the username and password to connect to the network. The login process is the same whether you connect by a wired or wireless connection.



Home network

You do not need a username and password for your home network. Everyone uses the same login details. To log in to the network, you need to know the network name and **network security key** for the router. This information is on a sticker on your router. The network security key works like a password.

When you switch on your computer, you will be shown a list of available networks. Your computer will detect your home network and any other networks close to you. It will detect your neighbour's network, if they are close enough, and any public hotspots.



When you click on the name of your network, you are asked to enter the network security key. Entering it correctly will connect you to the network. When you have entered the key once, your computer will remember it. If you tick the 'Connect automatically' box, you won't need to log on in future. Your computer will automatically connect to the network.

Solving network problems

When you are solving computer problems, you will be working with electrical equipment. Your priority is to stay safe. Always follow your school computer use safety rules. Always check with your parents before trying to deal with problems with your home network.

Stay safe

Always follow these safety rules.

- Switch off power points at the wall before connecting or disconnecting a mains electricity cable.
- Do not open the case of your computer or any other device.
- Never use force when connecting or disconnecting a cable or component.
- Make sure all cables are disconnected before attempting to move a device (for example, a printer).
- Do not lift heavy equipment on your own.
- If in doubt, STOP and ask for help.



Be creative

Create a poster to encourage people to act responsibly and safely when solving computer problems.

Make notes

When you are trying to solve network problems, record:

- the location of the computer
- the steps you have followed
- any error messages you see on-screen.

Username and password

If you are having problems logging on to a network, check you are using the correct username and password.

A password is case-sensitive. That means it matters if you use upper case letters instead of lower case ones. Check you haven't left 'Caps Lock' on.

If you have forgotten your password, use the 'Forgot password?' link to change your password. You will need an email address that a new password can be sent to. At school, you will need to ask your IT technician to change your password.

Wired networks

If you are having network problems with a wired connection, check that the cable connecting your computer to a network point is connected securely at both ends.

The connection socket on your computer will usually have a small green light. If the light is flashing, your computer is connected.

There are two things you can check if the light isn't flashing.

- The cable connecting your computer to the network may be faulty. Changing it for one you know is not faulty will solve the problem.
- The network point you are connected to may be faulty. Try another network point.



Only try to solve problems with your school network if your school IT policy allows it. Report the problem to the IT technician or your teacher and use another computer until the problem is fixed.

Wireless networks

If you cannot make a wireless connection to a network, check if you have wireless enabled on your computer. Find the Wi-Fi icon on the tool bar at the bottom of your screen. If it has a red cross on it, wireless on your computer may be turned off.

The picture on the right shows the wireless network symbol.



Click the Wi-Fi icon to open the wireless control box. You can turn wireless on by clicking the button marked 'Wi-Fi'.

If you are using a tablet or smartphone. you will find the same Wi-Fi icon on your device. If the icon is grey, wireless is turned off. Tap the icon to turn wireless on.

Home network

If you are having problems connecting to your home network, check:

- you are entering your password correctly
- that wireless is turned on
- whether other people in the house have a connection. If they do, the problem is likely to be with your device, not the router.

There are some basic checks you can carry out on your home network router.

- Is the router plugged into an electricity socket? Is the socket switched on?
- Check that any cables connected to the back of your router are secure.

If everything else fails, reset your router. Turn off the power switch or turn off the electricity socket. Wait 30 seconds then turn the router back on again. Do not press any button marked 'reset' on the router.



Activity

Write a list of instructions called 'How to connect to a network'.

Design the instructions for students who are about to start using computers, to help them study at school and at home.

Extra challenge

Work in a small group. Use your multimedia skills to create an audio or video guide to network connectivity. You can use a phone or recorder. Record yourself reading the instructions you wrote, or make a short video of someone connecting to the school network and explaining how to do it.

Test

1. List three ways to connect to a network.
2. What is a hotspot?
3. You are unable to connect your computer to your home network. List four problems that could be preventing you from connecting.
4. You want to connect a laptop to your school network. Why is a wireless connection better than a wired connection?

1.6 In the cloud

In this lesson

You will learn:

- about cloud storage
- about other services that are available in the cloud.

Spiral back

In Student Book 4 you learned about computer storage. You can save files using a storage device on your computer. Files can also be saved on a network storage device. You probably use network storage at school.

What is the cloud?

The **cloud** is a term used to describe the internet. The cloud is a metaphor for the internet. When you use a metaphor, you are saying that the internet is like a cloud, without using the word 'like'.

Saving a file to the cloud means saving the file to the internet.

Cloud storage

Everything you do 'in the cloud' involves data storage. When you save file in the cloud, it is stored on large storage devices. The storage devices are held in a **data centre**.

A data centre is a very large computer system. It contains many servers and storage devices. A data centre often has its own building.





Who provides cloud storage?

There are many providers of cloud storage. Your internet provider may offer cloud storage. Software providers like Microsoft and Google provide cloud storage. Microsoft's cloud storage is called OneDrive.

You often get free cloud storage when you buy a computer, or other devices like cameras and smartphones. Some companies specialise in providing cloud storage. Dropbox is an example.

Usually, a small amount of cloud storage space is provided free of charge. Businesses and people who have a lot of data can pay for more space if they need it.

Advantages of cloud storage

Here are the main advantages of cloud storage.

- You can access your files from anywhere you have an internet connection. If you save a file on your own computer, you can only access the file if you have the computer with you. If your file is in the cloud you can access it anywhere, using any computer.
- You can share your files with other people. For example, when you are working on team projects.
- Your work is safe. Cloud storage providers back up your files so that you don't have to do it. It is unlikely your data will be lost or stolen.

Other cloud services

Storage isn't the only reason for using the cloud. There are many other services.

Collaboration

As you saw in Student Book 7, collaboration means working with others.

Collaboration software makes working together easier. If files are stored in the cloud, any team member can work on them.

Collaboration software also lets people in different locations meet using web conferencing software. **Web conferencing** shows a video of the people taking part in the meeting. It also lets people show files on-screen and make notes. A web conference saves travel time for the people taking part.

Software applications

Many software applications (apps) work in the cloud, for example, Google Apps and Microsoft 365. The apps make it easy to work with other people. They make sharing documents and files easy.

Web hosting

If you want to build a website, you need a web server to store your web pages. A web server can be expensive, and most people do not have the skills to set up and run one.

Web hosting companies help people who want to set up a website. A web hosting service provides technical knowledge and web editing software. That makes creating a website easy.

Music, videos and games



Music and videos have been stored online for many years. When **digital** music was first stored online, people bought and downloaded the music. When downloading, you save a copy of the digital music file on your computer. When you have saved the copy, you can play the music on your computer, or transfer it onto a special MP3 player, such as an iPod.

Today, music and videos are often streamed to a computer or device like a smartphone. **Streaming** means that a music track is played as it is being downloaded to your computer. A copy of the file does not have to be saved before the music is played. Streaming is a cloud service.

Playing games in the cloud means that you can compete or collaborate with other players. This type of game is called a **multi-player game**. Multi-player games can be more interesting than single-player games. They often have social media tools included to allow players to chat about the game.

Advantages and disadvantages of cloud services

Here are the advantages.

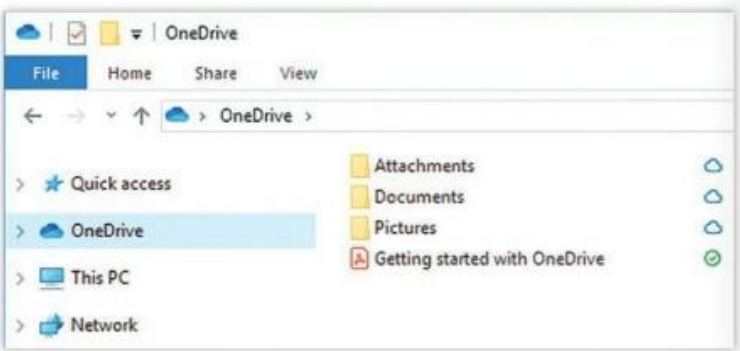
- Files are available anywhere and on any computer.
- Collaborating and sharing data is easy.
- Data is safe.
- Low cost to start with.
- Technical help is available from the company providing the service.

Here are the disadvantages.

- The service might not be exactly what you need.
- The owner of the service can change how it works.
- The owner of the service can increase costs.
- Can be expensive over time.

Saving to the cloud

When you save a file, you must choose where you save it. You can save it on a storage device on your computer. You can save it on a network storage device. You can save it to the cloud. The places will be listed for you when you save a file.



When you save a file to the cloud, a copy of the file is also saved to your own computer. This means you can open the file and work on it even if you have no internet connection.

You can choose not to make a copy of a file on your computer. The file will only be saved to the cloud. You will not be able to use the file if there is no internet connection. You can save large files like videos and photos to the cloud to save space on your computer.

Cloud storage uses icons to tell you how a file has been saved.

- A cloud status icon shows that the file is only saved in the cloud.
- A clock status icon tells you the file is saved to the cloud and your computer.
- An icon that looks like a person tells you the file has been shared with someone.



Activity

Your company is about to launch a new cloud data storage service.

Write an advert to attract as many new customers as possible. Make sure your advert highlights the advantages of cloud storage.

Extra challenge

Do you use any cloud services? For example, multi-player games, or streaming music and videos.

Make a list of the services you use. Which are your favourite services and why?

Is there anything you don't like about the cloud services you use?

Test

1. Describe three services that can be accessed in the cloud.
2. Where is data saved when it is saved to the cloud?
3. Explain why it is a good idea to store your data files in the cloud.
4. List three disadvantages of using cloud services.